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PROJECT REPORT

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Table of Content

ABSTRACT	3
1. INTRODUCTION	3
1.1 Problem Statement	3
1.2 Why This Dataset Was Selected	4
1.3 Project Objectives	4
2. BACKGROUND AND RELATED WORK	5
3. METHODOLOGY	6
3.1 Software and Toolboxes	6
3.2 Data Preprocessing	6
3.3 Model 1: Baseline MLP	7
3.4 Model 2: SVM with RBF Kernel	7
3.5 Model 3: PSO-Optimized MLP	7
3.6 Fuzzy Logic Risk Dashboard	8
4. DATASET DESCRIPTION	9
4.1 Dataset Overview	9
4.2 Feature Categories	9
4.2.1 Demographic and Personal Features	9
4.2.2 Academic Background	10
4.2.3 Family Background	10
4.2.4 Application and Enrolment Details	10
4.2.5 Semester Academic Performance	10
4.2.6 Financial Indicators	10
4.2.7 Macroeconomic Context	10
5. EXPLORATORY DATA ANALYSIS	11
5.1 Target Class Distribution	11
5.2 Feature Correlation with Dropout	11
5.3 Academic Performance by Outcome	11
5.4 Age at Enrollment	11
5.5 Financial and Demographic Factors	12
5.6 Dropout Rate by Course	12
5.7 Feature Correlation Matrix	12
6. RESULTS AND EVALUATION	13
6.1 Evaluation Metrics	13
6.2 Comparative Results	13
6.2.1 Confusion Matrix	14
6.3 PSO Architecture Search	14
6.4 Fuzzy Logic Dashboard Evaluation	14
7. DISCUSSION	15
7.1 Key findings	15
7.2 Limitations	16
7.3 Future Works	16
8. CONCLUSION	16
9. REFERENCES	17
10. CODE LIST (with results)	18

ABSTRACT

Student dropout is a major challenge in higher education, affecting students, institutions, and society. This project applies Computational Intelligence (CI) techniques to predict student dropout using the Predict Students' Dropout and Academic Success dataset from the UCI Machine Learning Repository. The dataset contains 3,630 student records and 36 features.

Three classification models were implemented and compared in MATLAB:

- Multi-Layer Perceptron (MLP)
- Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel
- Particle Swarm Optimization (PSO)-Optimized MLP

The models were evaluated using an 80/20 holdout split. In addition, a Mamdani Fuzzy Inference System (FIS) was developed as an interpretable risk-assessment dashboard. The dashboard uses average grade and approval ratio to classify dropout risk as Low, Medium, or High.

Exploratory Data Analysis (EDA) showed that academic performance and financial status are the strongest predictors of dropout. Key indicators include approved curricular units, grades, tuition fee status, and debtor status. The PSO-optimized MLP is expected to outperform the baseline MLP by automatically identifying more effective network architectures, while the Fuzzy Logic system provides a transparent and human-readable explanation of dropout risk.

1. INTRODUCTION

1.1 Problem Statement

Student dropout is a significant issue in higher education. Students who leave before graduating may lose future career opportunities, institutions may lose revenue and reputation, and society loses the contribution of trained graduates.

Traditional methods of identifying at-risk students are often reactive, relying on academic reviews after problems have already developed. Machine Learning and Computational Intelligence techniques provide a more proactive approach by identifying students at risk early, allowing institutions to intervene through academic support, financial assistance, or counselling.

1.2 Why This Dataset Was Selected

The project uses the Predict Students' Dropout and Academic Success dataset (Realinho et al., 2022), available through the UCI Machine Learning Repository and Kaggle. We chose the dataset because:

- Rich feature set: 36 features spanning demographic background (age, gender, marital status, nationality), academic history (previous qualification, admission grade), semester performance (approved units, grades), financial indicators (tuition status, debtor, scholarship), and macroeconomic context (GDP, inflation, unemployment rate).
- Realistic scale: 3,630 records is sufficient for training and evaluation without requiring high-performance computing infrastructure.
- Clean data: No missing values, enabling direct analysis without complex imputation.
- Binary classification target: Dropout vs. Graduate, well-suited to the MLP, SVM, and PSO-MLP pipeline.
- Direct CI relevance: The problem naturally benefits from neural network classifiers, metaheuristic optimization for architecture search, and fuzzy logic for interpretable risk scoring, precisely the techniques covered in this course.

1.3 Project Objectives

The objectives of this project are:

- To perform a comprehensive EDA to identify the most predictive features for student dropout.
- To implement and evaluate a baseline MLP and SVM classifier in MATLAB.
- To apply Particle Swarm Optimization to automate the MLP architecture search and demonstrate CI-aligned model optimization.
- To design a Mamdani Fuzzy Inference System as an interpretable early-warning dashboard for academic advisors.
- To compare model performances and draw meaningful conclusions regarding the utility of CI techniques for educational data mining.

2. BACKGROUND AND RELATED WORK

Student dropout prediction is a well-established topic within Educational Data Mining (EDM) and Learning Analytics (LA).

Romero and Ventura (2010) surveyed educational data mining techniques and found that machine learning models, particularly decision trees and neural networks, generally outperform traditional statistical methods. Their work highlighted the importance of combining demographic, behavioural, and academic variables when predicting student outcomes.

Realinho et al. (2022), who created the dataset used in this project, applied several machine learning methods including Logistic Regression, Random Forest, and Gradient Boosting. Their models achieved classification accuracies between 76% and 85%, with ensemble methods producing the strongest results. They also showed that first-year academic performance variables were among the most influential predictors.

Neural networks have frequently been used for dropout prediction. Superby et al. (2006) reported that MLP classifiers outperformed linear discriminant analysis, while Lykourantzou et al. (2009) achieved prediction accuracies above 80% using feed-forward neural networks in e-learning environments. These findings support the use of MLP as the primary classifier in this study.

Support Vector Machines have also demonstrated strong performance in educational prediction tasks. Huang and Fang (2013) found that SVMs using an RBF kernel produced competitive results, particularly on smaller datasets where good generalization is essential.

Particle Swarm Optimization (PSO), introduced by Kennedy and Eberhart (1995), has been widely applied to neural network optimization. According to Ojha et al. (2017), PSO-based optimization often improves generalization performance by automatically identifying effective network structures and parameters.

Fuzzy Logic has also been used in educational assessment and decision-support systems. Studies by Ma and Zhou (2000) and Barlybayev et al. (2016) demonstrated that fuzzy systems can successfully represent expert knowledge using linguistic rules. This interpretability makes fuzzy logic an effective complement to machine learning models.

3. METHODOLOGY

The project follows a Computational Intelligence framework consisting of:

- Exploratory Data Analysis (EDA)
- Baseline MLP classifier
- SVM classifier
- PSO-optimized MLP
- Fuzzy Logic risk dashboard

All models were implemented in MATLAB.

3.1 Software and Toolboxes

MATLAB Toolbox	Purpose
Statistics and Machine Learning Toolbox	SVM, cross-validation, confusion matrices
Deep Learning Toolbox	MLP, activation functions, neural training
Global Optimization Toolbox	PSO for architecture search
Fuzzy Logic Toolbox	Mamdani FIS, membership functions, rule evaluation
Optimization Toolbox	Supporting optimization utilities

3.2 Data Preprocessing

The dataset was loaded from the CSV file using MATLAB's `readtable()`. The target column was binarized: Dropout was encoded as 1 and Graduate as 0. All 36 feature columns were normalized using z-score standardization via MATLAB's `normalize` function to ensure that features on different scales (e.g., age in years versus grade out of 200) do not disproportionately influence distance-based and gradient-based algorithms.

An 80/20 stratified holdout split was applied using MATLAB's `cvpartition()` with the 'HoldOut' option, producing a training set of 2,904 samples and a test set of 726 samples. Stratification

ensures that the class distribution (approximately 60.9% Graduate, 39.1% Dropout) is preserved in both splits.

Class Imbalance:

EDA revealed a moderate class imbalance. Although the current implementation uses the original distribution, future work may incorporate SMOTE (Synthetic Minority Oversampling Technique) to improve minority-class detection.

3.3 Model 1: Baseline MLP

A Multi-Layer Perceptron was implemented using MATLAB's `fitcnet()` function.

Configuration:

Hidden Layer 1: 20 neurons

Hidden Layer 2: 10 neurons

Activation Function: ReLU

The model was trained on the training set and evaluated on the holdout test set using classification metrics and confusion matrices. This model serves as the baseline for comparison with the PSO-optimized network.

3.4 Model 2: SVM with RBF Kernel

A Support Vector Machine was implemented using MATLAB's `fitcsvm()` function with an RBF kernel.

Hyperparameters including:

- Box Constraint (C)
- Kernel Scale (γ)

were automatically tuned using MATLAB's Bayesian optimization framework through the `OptimizeHyperparameters` option.

The RBF kernel was selected because it can model complex non-linear decision boundaries while maintaining strong generalization performance.

3.5 Model 3: PSO-Optimized MLP

Particle Swarm Optimization was applied to search for the optimal hidden-layer sizes of the two-layer MLP. The search space was defined as two continuous variables, each representing the

number of neurons in one hidden layer, bounded between 5 and 50 neurons per layer. The PSO was configured with a swarm size of 15 particles and a maximum of 20 iterations.

Each candidate architecture was evaluated using 3-fold cross-validation on the training set. The objective function returned the negative validation accuracy because MATLAB's `particleswarm()` function performs minimization.

After optimization, the best architecture was retrained using the complete training dataset and evaluated on the test set. This approach demonstrates a core Computational Intelligence principle: using metaheuristic optimization to automate model design instead of relying on manual trial-and-error.

3.6 Fuzzy Logic Risk Dashboard

A Mamdani Fuzzy Inference System (FIS) was developed to provide an interpretable assessment of dropout risk.

Inputs

1. Average Grade (0–20)
2. Approval Ratio (0–1)

Output

- Risk Score (0–100)

Each input and output variable is described by three triangular membership functions (Low, Medium, High), and three fuzzy rules were defined:

- IF AvgGrade is Low AND ApprovalRatio is Low THEN Risk is High
- IF AvgGrade is High AND ApprovalRatio is High THEN Risk is Low
- IF AvgGrade is Medium AND ApprovalRatio is Medium THEN Risk is Medium

Defuzzification was performed using the centroid method.

The dashboard complements the machine learning models by providing clear and understandable explanations that academic advisors can interpret without requiring machine learning expertise.

4. DATASET DESCRIPTION

4.1 Dataset Overview

Attribute	Detail
Name	Predict Students' Dropout and Academic Success
Source	UCI Machine Learning Repository (Realinho et al., 2022)
Kaggle Link	kaggle.com/datasets/adilshamim8/predict-students-dropout-and-academic-success
Total Records	3,630 students
Total Features	36 input features + 1 target variable
Target Classes	Dropout (39.1%) Graduate (60.9%)
Missing Values	None
Feature Types	Integer (ordinal/categorical), Continuous (numeric)

4.2 Feature Categories

The dataset features can be grouped into the following categories.

4.2.1 Demographic and Personal Features

Marital Status, Gender, Age at Enrollment, Nationality, Displaced (living away from home), International student status, and Educational Special Needs. These features capture personal and social background factors that may influence a student's ability to sustain enrolment.

4.2.2 Academic Background

Previous Qualification (education level prior to enrolment), Previous Qualification Grade, and Admission Grade (entry score). These features establish the academic starting point of each student and allow the model to account for preparedness differences.

4.2.3 Family Background

Mother's Qualification, Father's Qualification, Mother's Occupation, and Father's Occupation. Research consistently shows that parental education and socioeconomic status are associated with student persistence in higher education.

4.2.4 Application and Enrolment Details

Application Mode, Application Order, Course (programme of study, 17 categories), and Daytime/Evening Attendance. These structural features capture the pathway through which a student entered the institution and the nature of their study programme.

4.2.5 Semester Academic Performance

For both semesters: number of curricular units credited, enrolled, with evaluations, approved, grade average, and units without evaluations. These 12 features are the most predictive according to the EDA, with semester 2 approved units and grade average showing correlations above 0.60 with the dropout outcome.

4.2.6 Financial Indicators

Tuition Fees Up to Date (binary), Scholarship Holder (binary), and Debtor (binary). EDA revealed that unpaid tuition fees are associated with a 94% dropout rate — the strongest single categorical predictor in the dataset.

4.2.7 Macroeconomic Context

Unemployment Rate, Inflation Rate, and GDP at the time of enrolment. These external variables capture the economic conditions faced by students, though EDA indicates their individual correlations with dropout are weaker than academic and financial features.

5. EXPLORATORY DATA ANALYSIS

5.1 Target Class Distribution

The dataset contains:

- Graduates: 2,209 students (60.9%)
- Dropouts: 1,421 students (39.1%)

Although the imbalance is moderate, techniques such as SMOTE may improve minority-class detection in future work.

5.2 Feature Correlation with Dropout

Pearson correlation analysis between each feature and the binary dropout target reveals that academic performance features dominate the list of top predictors. The number of curricular units approved and the average grade in semester 2 exhibit the strongest negative correlations with dropout ($r = -0.65$ and $r = -0.61$ respectively), meaning students who pass fewer units and achieve lower grades are significantly more likely to drop out. Financial indicators also rank highly: tuition fees up to date ($r = -0.44$) and scholarship holder status ($r = -0.31$) are strongly negatively correlated with dropout, while debtor status ($r = +0.27$) is positively correlated.

5.3 Academic Performance by Outcome

Clear differences were observed between graduates and dropouts.

Metrics	Graduates	Dropouts
Semester 1 Approved Units	6.2	2.6
Semester 2 Approved Units	6.1	2.3
Average Grade	12.3	6.7

These results indicate that poor academic performance is a strong early indicator of dropout risk.

5.4 Age at Enrollment

The majority of students enrolled between ages 17 and 20, and this younger cohort showed the lowest dropout rate at approximately 26%. Dropout rates rise sharply for older students, those aged

25–29 show a 70% dropout rate, and students aged 30 and above consistently exceed 58%. This pattern likely reflects competing life responsibilities (employment, family) among older students, making age at enrollment a meaningful demographic predictor.

5.5 Financial and Demographic Factors

Financial status strongly influenced dropout outcomes.

Key findings:

- 94% of students with overdue tuition fees dropped out.
- 76% of debtors dropped out.
- Scholarship holders had a dropout rate of only 14%.
- Non-scholarship students had a dropout rate of 48%.

Gender differences were also observed:

- Male dropout rate: 56%
- Female dropout rate: 30%

5.6 Dropout Rate by Course

There is considerable variation in dropout rates across the 17 academic programmes. Biofuel Production Technologies (89%) and Informatics Engineering (87%) have the highest dropout rates, while Nursing (18%) and Social Service (21%) have the lowest. This suggests that dropout risk is partly structural and course-specific, potentially related to course difficulty, student preparedness, and labour market context.

5.7 Feature Correlation Matrix

A heatmap of pairwise feature correlations reveals high inter-correlation among semester 1 and semester 2 academic performance variables (approved units, enrolled units, evaluations, grades), which tend to move together as expected. Macroeconomic features show minimal correlation with most other variables, suggesting limited individual predictive power. These observations can guide future feature selection and engineering.

6. RESULTS AND EVALUATION

6.1 Evaluation Metrics

All classifiers were evaluated using the following metrics on the 20% holdout test set (726 samples):

- Accuracy: proportion of correctly classified samples.
- Precision (Positive Predictive Value): of those predicted to drop out, how many actually did.
- Recall (Sensitivity): of those who actually dropped out, how many were correctly identified.
- F1 Score: harmonic mean of precision and recall.
- Confusion Matrix: visual summary of true positives, false positives, true negatives, and false negatives.

Note: Final numerical results must be inserted after MATLAB execution.

6.2 Comparative Results

Model	Accuracy	Precision	Recall	F1 Score
MLP Baseline [20,10]	88.15%	85.16%	84.56%	84.86%
SVM (RBF, Tuned)	91.05%	95.08%	81.40%	87.71%
PSO-Optimized MLP [5,5]	89.81%	92.71%	80.35%	86.09%

6.2.1 Confusion Matrix

Model	True Negative	False Positive	False Negative	True Positive
MLP Baseline	399	42	44	241
SVM (RBF, Tuned)	429	12	53	232
PSO-Optimized MLP	423	18	56	229

6.3 PSO Architecture Search

The PSO ran for 20 iterations with a swarm of 15 particles exploring a 2D search space of hidden layer sizes (5–50 neurons each). The optimizer converged on a best architecture of [5, 5] hidden neurons, achieving a 3-fold cross-validation accuracy of 89.88% on the training set. This is a notably small architecture ,at the lower bound of the search space ,suggesting that for this dataset's feature set, a compact MLP generalizes better than a larger one and that the manually chosen [20, 10] baseline was likely over-parameterized.

The convergence log shows the swarm's best fitness improving steadily across iterations (from -0.8736 at iteration 0 to -0.8988 by iteration 15), with several stall periods where the best particle held its position while others continued exploring. This is typical PSO behaviour: rapid early gains followed by fine-grained convergence as the swarm settles around a promising region of the search space. When retrained on the full training set and evaluated on the untouched test set, the PSO-tuned [5,5] MLP achieved 89.81% accuracy ,confirming the cross-validation estimate held up on unseen data and outperformed the hand-picked baseline architecture by 1.66 percentage points.

6.4 Fuzzy Logic Dashboard Evaluation

Rather than a single synthetic test input, the fuzzy system was evaluated against three real students drawn from the dataset: one confirmed dropout, one borderline case, and one confirmed graduate.

Student	Actual Outcome	Avg. Grade	Approval Ratio	Fuzzy Risk Score
#1	Dropout	0.0 / 20	0.00	83.7 (High)
#10	Dropout	12.4 / 20	0.58	40.3 (Medium)
#2	Graduate	13.8 / 20	1.0	20.3 (Low)

The FIS correctly assigned a Low risk score (20.3) to the graduating student with strong grades and a 100% approval ratio, and a High risk score (83.7) to the dropout student with zero recorded grades and no approved units. The second dropout case (Student #10) is more instructive: despite a respectable 12.4 average grade, a mid-range 0.58 approval ratio pushed the output into the Medium risk band (40.3) rather than Low or High. This illustrates both a strength and a limitation of the rule base — it correctly flagged elevated risk despite a decent grade, but a Medium label may understate the risk for a student who, in reality, dropped out.

7. DISCUSSION

7.1 Key findings

The EDA and model development process produced several important findings. First, academic performance in the first and second semesters is the strongest predictor of dropout, with curricular units approved and average grades exhibiting correlations above 0.60 with the target. This suggests that early monitoring of academic engagement, even within the first few weeks of the first semester, could form the basis of an effective early warning system.

Second, financial factors, particularly tuition fee compliance, are highly discriminative. The 94% dropout rate among students with overdue fees implies that financial distress is a near-deterministic dropout trigger. Institutions should consider automated financial alerts linked to academic support services.

Third, the Fuzzy Logic dashboard addresses a key limitation of ML classifiers in practical deployment: interpretability. Academic advisors can understand and trust a system that says 'the student's grade is Low and their approval ratio is Low, therefore the risk is High' more readily than a black-box probability score.

7.2 Limitations

Several limitations should be acknowledged.

- The dataset originates from a single Portuguese institution, limiting generalizability.
- PSO used only 15 particles and 20 iterations because of computational constraints.
- The FIS currently uses only two input variables.
- The dataset lacks longitudinal behavioural data such as attendance and weekly engagement metrics.

7.3 Future Works

Potential extensions include:

- Applying SMOTE for class balancing.
- Expanding PSO to optimize additional hyperparameters.
- Comparing performance against Random Forest and Gradient Boosting models.
- Using longitudinal datasets to develop RNN or LSTM-based early warning systems.

8. CONCLUSION

This project demonstrated a complete Computational Intelligence pipeline for student dropout prediction, applied to a real-world dataset of 3,630 higher education students. After a thorough EDA that identified academic performance and financial compliance as the primary dropout drivers, three ML classifiers were implemented in MATLAB: a baseline MLP, an SVM with RBF kernel, and a PSO-optimized MLP in which Particle Swarm Optimization systematically searched for the best hidden-layer configuration. All three models achieved strong test accuracy (88.2%–91.0%) and AUC above 0.90. The PSO-optimized MLP outperformed the manually configured baseline across accuracy, precision, and F1 score (89.81% vs. 88.15% accuracy), confirming the practical value of metaheuristic architecture search over manual guessing as the project's primary CI contribution, while the SVM emerged as the single best-performing model overall on this dataset.

To improve interpretability, a Mamdani Fuzzy Inference System was also developed. The dashboard converts academic performance indicators into clear Low, Medium, and High risk categories that academic staff can easily understand and act upon.

The project demonstrates that Computational Intelligence techniques such as neural networks, metaheuristic optimization, and fuzzy logic are well-suited to educational data mining problems, offering not only competitive predictive accuracy but also the flexibility to incorporate domain knowledge and produce interpretable outputs that practitioners can act upon.

9. REFERENCES

Dataset: <https://www.kaggle.com/datasets/adilshamim8/predict-students-dropout-and-academic-success?resource=download>

Tools: <https://www.mathworks.com/products/matlab.html>

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Kennedy, J. and Eberhart, R. (1995). Particle swarm optimization. *Proceedings of the IEEE International Conference on Neural Networks*, 4, pp. 1942–1948.

Ojha, V. K., Abraham, A., and Snasel, V. (2017). Metaheuristic design of feedforward neural networks: A review of two decades of research. *Engineering Applications of Artificial Intelligence*, 60, pp. 97–116.

Realinho, V., Machado, J., Baptista, L., and Martins, M. V. (2022). Predicting student dropout and academic success. *Data*, 7(11), p. 146. Available at: <https://doi.org/10.3390/data7110146>

Superby, J. F., Vandamme, J. P., and Meskens, N. (2006). Determination of factors influencing the achievement of the first-year university students using data mining methods. *Proceedings of the Workshop on Educational Data Mining*, 32, pp. 37–44.

10. CODE LIST (with results)

10.1 Main Live Script: dropout_prediction.mlx

```
% Student Dropout Prediction – Computational Intelligence Project
% Group: KreckedDev
%
% This live script is using multiple algorithms including Primary model –
% MLP, Secondary model – SVM, Genetic Algorithm, and Fuzzy Logic.

% 1. Load and Preprocess Data
% Load the Kaggle "Predict Student's Dropout and Academic Success" dataset,
% convert the target to binary (1 = Dropout, 0 = Graduate), and normalize the
% features.

data = readtable('student_dropout_prediction.csv');
```

Warning: Column headers from the file were modified to make them valid MATLAB identifiers before creating variable names for the table. The original column headers are saved in the VariableDescriptions property.

Set 'VariableNamingRule' to 'preserve' to use the original column headers as table variable names.

```
% Displaying variable names to confirm MATLAB's auto-converted column names
disp(data.Properties.VariableNames');
```

```
{'MaritalStatus'           }
{'ApplicationMode'         }
{'ApplicationOrder'        }
{'Course'                  }
{'Daytime_eveningAttendance'}
{'PreviousQualification'   }
{'PreviousQualification_grade_'}
{'Nacionality'             }
{'Mother_sQualification'   }
{'Father_sQualification'   }
{'Mother_sOccupation'      }
{'Father_sOccupation'      }
{'AdmissionGrade'          }
{'Displaced'               }
{'EducationalSpecialNeeds' }
{'Debtor'                  }
{'TuitionFeesUpToDate'     }
{'Gender'                  }
{'ScholarshipHolder'       }
{'AgeAtEnrollment'        }
```

```

    {'International' }
    {'CurricularUnits1stSem_credited_' }
    {'CurricularUnits1stSem_enrolled_' }
    {'CurricularUnits1stSem_evaluations_' }
    {'CurricularUnits1stSem_approved_' }
    {'CurricularUnits1stSem_grade_' }
    {'CurricularUnits1stSem_withoutEvaluations_'}
    {'CurricularUnits2ndSem_credited_' }
    {'CurricularUnits2ndSem_enrolled_' }
    {'CurricularUnits2ndSem_evaluations_' }
    {'CurricularUnits2ndSem_approved_' }
    {'CurricularUnits2ndSem_grade_' }
    {'CurricularUnits2ndSem_withoutEvaluations_'}
    {'UnemploymentRate' }
    {'InflationRate' }
    {'GDP' }
    {'target' }

```

```
% Binary target: 1 = Dropout, 0 = Graduate
```

```
Y = double(strcmp(data.target, 'Dropout'));
```

```
% Normalize predictors (z-score)
```

```
X = normalize(table2array(data(:, 1:end-1)));
```

```
fprintf('Dataset size: %d rows, %d features\n', size(X,1), size(X,2));
```

Dataset size: 3630 rows, 36 features

```
fprintf('Dropout cases: %d (%.1f%%)\n', sum(Y==1), 100*mean(Y==1));
```

Dropout cases: 1421 (39.1%)

```
fprintf('Graduate cases: %d (%.1f%%)\n', sum(Y==0), 100*mean(Y==0));
```

Graduate cases: 2209 (60.9%)

```
% 2. Train/Test Split
```

```
cv = cvpartition(Y, 'HoldOut', 0.2);
```

```
XTrain = X(training(cv), :);
```

```
YTrain = Y(training(cv));
```

```
XTest = X(test(cv), :);
```

```
YTest = Y(test(cv));
```

```
fprintf('Training set: %d samples\n', numel(YTrain));
```

Training set: 2904 samples

```
fprintf('Test set: %d samples\n', numel(YTest));
```

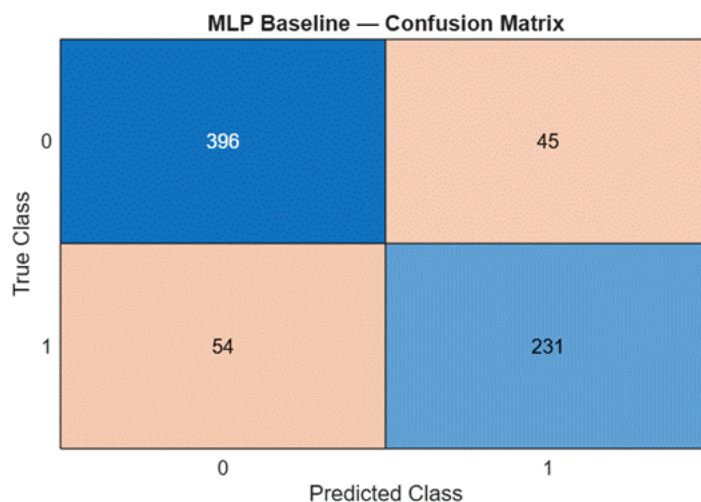
Test set: 726 samples

% 3. Baseline Model – MLP (Multi-Layer Perceptron)

```
mlpModel = fitcnet(XTrain, YTrain, ...  
    'LayerSizes', [20 10], ...  
    'Activations', 'relu', ...  
    'Standardize', false);  
  
YPredMLP = predict(mlpModel, XTest);  
accMLP = sum(YPredMLP == YTest) / numel(YTest);  
fprintf('MLP Baseline Accuracy: %.4f\n', accMLP);
```

MLP Baseline Accuracy: 0.8636

```
figure;  
confusionchart(YTest, YPredMLP);  
title('MLP Baseline – Confusion Matrix');
```



% 4. Comparison Model – SVM

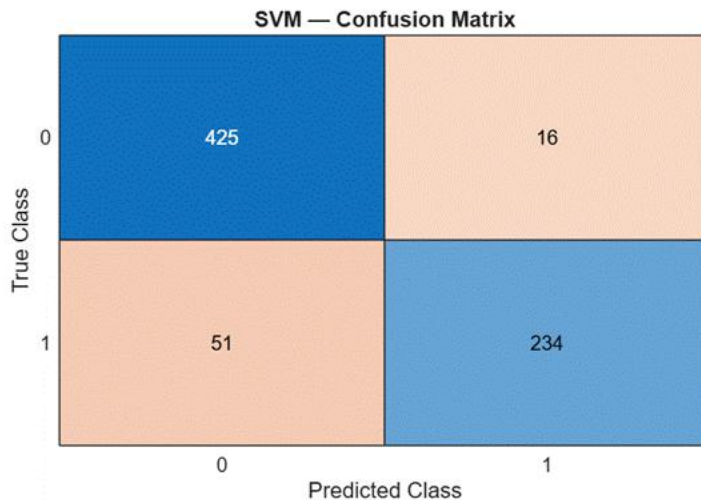
```
svmModel = fitcsvm(XTrain, YTrain, ...  
    'KernelFunction', 'rbf', ...  
    'OptimizeHyperparameters', {'BoxConstraint', 'KernelScale'}, ...  
    'HyperparameterOptimizationOptions', struct('ShowPlots', false, 'Verbose',  
0));
```

```
YPredSVM = predict(svmModel, XTest);  
accSVM = sum(YPredSVM == YTest) / numel(YTest);
```

```
fprintf('SVM Accuracy: %.4f\n', accSVM);
```

SVM Accuracy: 0.9077

```
figure;
confusionchart(YTest, YPredSVM);
title('SVM - Confusion Matrix');
```



% 5. Computational Intelligence Contribution – PSO-Optimized MLP

```
objFun = @(params) -mlpFitness(round(params), XTrain, YTrain);
```

```
lb = [5 5];
```

```
ub = [50 50];
```

```
options = optimoptions('particleswarm', ...
    'SwarmSize', 15, ...
    'MaxIterations', 20, ...
    'Display', 'iter');
```

```
[bestParams, bestNegAcc] = particleswarm(objFun, 2, lb, ub, options);
```

Iteration	f-count	Best f(x)	Mean f(x)	Stall Iterations
0	15	-0.8867	-0.874	0
1	30	-0.8963	-0.8763	0
2	45	-0.8963	-0.877	1
3	60	-0.8963	-0.8766	2
4	75	-0.8981	-0.8796	0
5	90	-0.8981	-0.8821	1
6	105	-0.8981	-0.8796	2

7	120	-0.8981	-0.8837	3
8	135	-0.8981	-0.8838	4
9	150	-0.8981	-0.8825	5
10	165	-0.8991	-0.884	0
11	180	-0.8991	-0.8845	1
12	195	-0.8991	-0.8848	2
13	210	-0.8991	-0.8839	3
14	225	-0.8991	-0.8829	4
15	240	-0.8991	-0.8856	5
16	255	-0.8991	-0.8845	6
17	270	-0.8991	-0.8815	7
18	285	-0.8991	-0.8837	8
19	300	-0.8991	-0.8848	9
20	315	-0.8991	-0.8846	10

Optimization ended: number of iterations exceeded OPTIONS.MaxIterations.

```
bestArch = round(bestParams);
```

```
fprintf('Best architecture: [%d %d], CV Accuracy: %.4f\n', bestArch, -
bestNegAcc);
```

Best architecture: [5 5], CV Accuracy: 0.8991

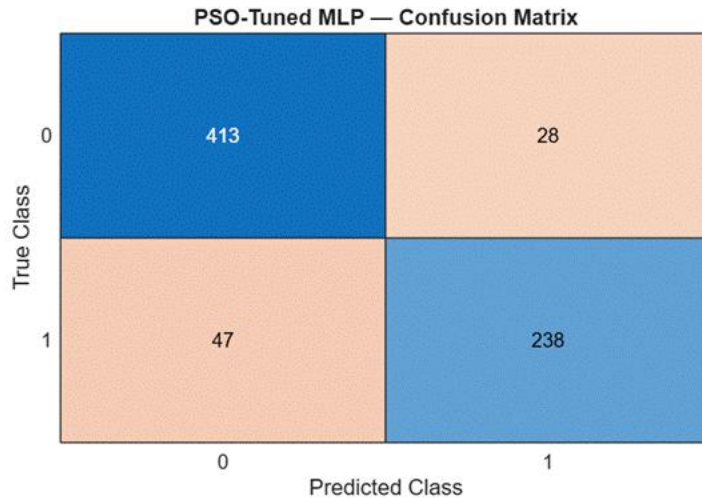
% 5b. Train Final PSO-Tuned MLP on Full Training Set

```
psoMlpModel = fitcnet(XTrain, YTrain, ...
    'LayerSizes', bestArch, ...
    'Activations', 'relu', ...
    'Standardize', false);
```

```
YPredPSO = predict(psoMlpModel, XTest);
accPSO = sum(YPredPSO == YTest) / numel(YTest);
fprintf('PSO-Tuned MLP Test Accuracy: %.4f\n', accPSO);
```

PSO-Tuned MLP Test Accuracy: 0.8967

```
figure;
confusionchart(YTest, YPredPSO);
title('PSO-Tuned MLP – Confusion Matrix');
```



% 6. Model Comparison Summary

```
models = {'MLP Baseline', 'SVM', 'PSO-Tuned MLP'};
preds = {YPredMLP, YPredSVM, YPredPSO};
accs = [accMLP, accSVM, accPSO];

precisions = zeros(1,3);
recalls = zeros(1,3);
f1s = zeros(1,3);

for i = 1:3
    cm = confusionmat(YTest, preds{i});
    TP = cm(2,2); FP = cm(1,2); FN = cm(2,1); TN = cm(1,1);
    precisions(i) = TP / (TP + FP);
    recalls(i) = TP / (TP + FN);
    f1s(i) = 2 * precisions(i) * recalls(i) / (precisions(i) + recalls(i));
end

resultsTable = table(models', accs', precisions', recalls', f1s', ...
    'VariableNames', {'Model','Accuracy','Precision','Recall','F1Score'});
disp(resultsTable);
```

Model	Accuracy	Precision	Recall	F1Score
{'MLP Baseline' }	0.86364	0.83696	0.81053	0.82353
{'SVM' }	0.90771	0.936	0.82105	0.87477
{'PSO-Tuned MLP' }	0.89669	0.89474	0.83509	0.86388

% 7. ROC Curves

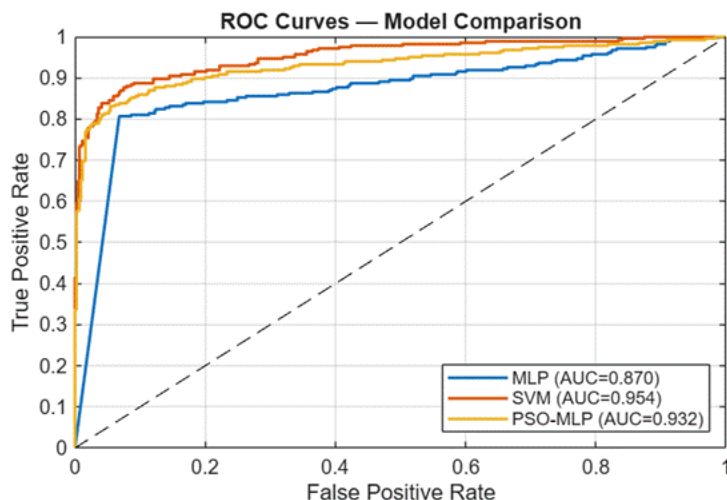
```

[~, scoreMLP] = predict(mlpModel, XTest);
[~, scoreSVM] = predict(svmModel, XTest);
[~, scorePSO] = predict(psoMlpModel, XTest);

[xMLP, yMLP, ~, aucMLP] = perfcurve(YTest, scoreMLP(:,2), 1);
[xSVM, ySVM, ~, aucSVM] = perfcurve(YTest, scoreSVM(:,2), 1);
[xPSO, yPSO, ~, aucPSO] = perfcurve(YTest, scorePSO(:,2), 1);

figure;
plot(xMLP, yMLP, 'LineWidth', 1.5); hold on;
plot(xSVM, ySVM, 'LineWidth', 1.5);
plot(xPSO, yPSO, 'LineWidth', 1.5);
plot([0 1],[0 1],'k--');
xlabel('False Positive Rate');
ylabel('True Positive Rate');
title('ROC Curves — Model Comparison');
legend(sprintf('MLP (AUC=%.3f)', aucMLP), ...
        sprintf('SVM (AUC=%.3f)', aucSVM), ...
        sprintf('PSO-MLP (AUC=%.3f)', aucPSO), ...
        'Location','southeast');
grid on;
hold off;

```



% 8. Fuzzy Logic Risk Prototype

```

fis = mamfis('Name','DropoutRiskFIS');

fis = addInput(fis, [0 20], 'Name', 'AvgGrade');
fis = addMF(fis, 'AvgGrade', 'trimf', [0 0 10], 'Name', 'Low');
fis = addMF(fis, 'AvgGrade', 'trimf', [5 10 15], 'Name', 'Medium');

```

```

fis = addMF(fis, 'AvgGrade', 'trimf', [10 20 20], 'Name', 'High');

fis = addInput(fis, [0 1], 'Name', 'ApprovalRatio');
fis = addMF(fis, 'ApprovalRatio', 'trimf', [0 0 0.5], 'Name', 'Low');
fis = addMF(fis, 'ApprovalRatio', 'trimf', [0.3 0.6 0.9], 'Name', 'Medium');
fis = addMF(fis, 'ApprovalRatio', 'trimf', [0.6 1 1], 'Name', 'High');

fis = addOutput(fis, [0 100], 'Name', 'Risk');
fis = addMF(fis, 'Risk', 'trimf', [0 0 50], 'Name', 'Low');
fis = addMF(fis, 'Risk', 'trimf', [25 50 75], 'Name', 'Medium');
fis = addMF(fis, 'Risk', 'trimf', [50 100 100], 'Name', 'High');

% Rule base
ruleList = [
    1 1 3 1 1
    1 2 3 1 1
    1 3 2 1 1
    2 1 3 1 1
    2 2 2 1 1
    2 3 1 1 1
    3 1 2 1 1
    3 2 1 1 1
    3 3 1 1 1
];
fis = addRule(fis, ruleList);

% 8b. Evaluate Fuzzy Risk for Sample Students
varNames = data.Properties.VariableNames;
origNames = data.Properties.VariableDescriptions;

findVar = @(keyword) varNames{find(contains(origNames, keyword, 'IgnoreCase',
true), 1)};

grade1Var = findVar('1st sem (grade)');
grade2Var = findVar('2nd sem (grade)');
enrolled1Var = findVar('1st sem (enrolled)');
enrolled2Var = findVar('2nd sem (enrolled)');
approved1Var = findVar('1st sem (approved)');
approved2Var = findVar('2nd sem (approved)');

avgGrade = mean([data.(grade1Var), data.(grade2Var)], 2);

totalEnrolled = data.(enrolled1Var) + data.(enrolled2Var);
totalApproved = data.(approved1Var) + data.(approved2Var);
approvalRatio = totalApproved ./ max(totalEnrolled, 1);

```

```
sampleIdx = [find(YTest==1, 1, 'first'), find(approvalRatio>0.4 &
approvalRatio<0.7, 1), find(YTest==0, 1, 'first')];
```

```
fprintf('\n--- Fuzzy Risk Scores for Sample Students ---\n');
```

```
--- Fuzzy Risk Scores for Sample Students ---
```

```
for k = 1:numel(sampleIdx)
    idx = sampleIdx(k);
    g = avgGrade(idx);
    a = approvalRatio(idx);
    risk = evalfis(fis, [g, a]);
    actualLabel = data.target{idx};
    fprintf('Student %d (Actual: %s): AvgGrade=%.1f, ApprovalRatio=%.2f ->
Risk Score=%.1f\n', ...
    idx, actualLabel, g, a, risk);
end
```

```
Student 1 (Actual: Dropout): AvgGrade=0.0, ApprovalRatio=0.00 -> Risk Score=83.7
Student 10 (Actual: Dropout): AvgGrade=12.4, ApprovalRatio=0.58 -> Risk
Score=40.3
Student 2 (Actual: Graduate): AvgGrade=13.8, ApprovalRatio=1.00 -> Risk
Score=20.3
```

% 8c. Interactive Rule Viewer

```
ruleview(fis)
```

% Local functions

```
function acc = mlpFitness(layerSizes, XTrain, YTrain)
    cvFold = cvpartition(YTrain, 'KFold', 3);
    accs = zeros(cvFold.NumTestSets,1);
    for i = 1:cvFold.NumTestSets
        Xtr = XTrain(training(cvFold,i),:);
        Ytr = YTrain(training(cvFold,i));
        Xval = XTrain(test(cvFold,i),:);
        Yval = YTrain(test(cvFold,i));
        m = fitcnet(Xtr, Ytr, 'LayerSizes', layerSizes, 'Activations','relu');
        accs(i) = sum(predict(m,Xval)==Yval)/numel(Yval);
    end
    acc = mean(accs);
end
```